

GNU nano 7.2

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;      // Global variable
static int static_var = 20; // Static variable

void code_function()
{
    printf("Inside code segment\n");
}

int main()
{
    int stack_var = 30;      // Stack variable
    int *heap_var = malloc(sizeof(int)); // Heap variable

    *heap_var = 40;

    printf("Process ID: %d\n", getpid());

    printf("Code address   : %p\n", (void *)code_function);
    printf("Global address  : %p\n", (void *)&global_var);
    printf("Static address   : %p\n", (void *)&static_var);
    printf("Heap address    : %p\n", (void *)heap_var);
    printf("Stack address   : %p\n", (void *)&stack_var);

    printf("\nPress Enter to exit...\n");
    getchar();

    free(heap_var);

    return 0;
}
```

```

compilation terminated.
sravya@Sravya1955:~/os$ gcc memory1.c -o memory1
sravya@Sravya1955:~/os$ ./memory1
Process ID: 490
Code address      : 0x63449a1bf209
Global address    : 0x63449a1c2010
Static address    : 0x63449a1c2014
Heap address      : 0x6344d437a2a0
Stack address     : 0x7ffec3ca6f6c

Press Enter to exit...
exit
sravya@Sravya1955:~/os$ nano memory1.c
sravya@Sravya1955:~/os$ |

```

```

sravya@Sravya1955: ~
sravya@Sravya1955:~$ cat /proc/490/maps
63449a1be000-63449a1bf000 r--p 00000000 08:30 45151 /home/sravya/os/memory1
63449a1bf000-63449a1c0000 r-xp 00001000 08:30 45151 /home/sravya/os/memory1
63449a1c0000-63449a1c1000 r--p 00002000 08:30 45151 /home/sravya/os/memory1
63449a1c1000-63449a1c2000 r--p 00002000 08:30 45151 /home/sravya/os/memory1
63449a1c2000-63449a1c3000 rw-p 00003000 08:30 45151 /home/sravya/os/memory1
6344d437a000-6344d439b000 rw-p 00000000 00:00 0 [heap]
765b42200000-765b42228000 r--p 00000000 08:30 47955 /usr/lib/x86_64-linux-gnu/libc.so.6
765b42228000-765b423b0000 r-xp 00028000 08:30 47955 /usr/lib/x86_64-linux-gnu/libc.so.6
765b423b0000-765b423ff000 r--p 001b0000 08:30 47955 /usr/lib/x86_64-linux-gnu/libc.so.6
765b423ff000-765b42403000 r--p 001fe000 08:30 47955 /usr/lib/x86_64-linux-gnu/libc.so.6
765b42403000-765b42405000 rw-p 00202000 08:30 47955 /usr/lib/x86_64-linux-gnu/libc.so.6
765b42405000-765b42412000 rw-p 00000000 00:00 0
765b42583000-765b42586000 rw-p 00000000 00:00 0
765b4258b000-765b4258d000 rw-p 00000000 00:00 0
765b4258d000-765b42591000 r--p 00000000 00:00 0 [vvar]
765b42591000-765b42593000 r--p 00000000 00:00 0 [vvar_vclock]
765b42593000-765b42595000 r-xp 00000000 00:00 0 [vdso]
765b42595000-765b42596000 r--p 00000000 08:30 47952 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
765b42596000-765b425c1000 r-xp 00001000 08:30 47952 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
765b425c1000-765b425cb000 r--p 0002c000 08:30 47952 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
765b425cb000-765b425cd000 r--p 00036000 08:30 47952 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
765b425cd000-765b425cf000 rw-p 00038000 08:30 47952 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
765b425cf000-765b425d0000 rw-p 00000000 00:00 0 [stack]

```